DO NOW · AO1 REVISION

Conservation of energy, power and efficiency

Energy
Paper 1 · Combined and Separate
22 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**Define power.**	The rate at which energy is transferred (or work is done).	☐
2 ★	**State the standard unit of power.**	Watt (W)	☐
3	**State the standard unit of time.**	Second (s)	☐
4	**State a unit that is equivalent to 1 W.**	1 J/s	☐
5	**State the term for energy transferred to stores that are not useful.**	Wasted energy	☐
6 ★	**State how unwanted energy transfers can be reduced.**	By lubrication and by thermal insulation.	☐
7	**Lubricating moving parts reduces unwanted energy transfers to which energy store?**	The thermal energy store	☐
8 ★	**Describe what it means for a device to be efficient.**	Most of the input energy is transferred usefully; little is wasted.	☐
9	**State the unit of efficiency.**	It has no unit (it is a ratio).	☐
10	**State how the efficiency of a mechanical energy transfer can be increased.**	By lubricating the moving parts.	☐
11	**State how the efficiency of a heating energy transfer can be increased.**	By adding thermal insulation.	☐
12	**Torches A and B each have an input power of 50 W. A has a useful output of 40 W and B of 45 W. Which is more efficient, and why?**	Torch B. Its efficiency is higher: 90%, against 80% for A.	☐

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	QUESTION	ANSWER	✓
13	**Heaters A and B each have a useful output of 2000 W. A has an input power of 2500 W and B of 2600 W. Which is more efficient, and why?**	Heater A. Its efficiency is higher: 80%, against 77% for B.	☐
14	**State the equation linking power, energy transferred and time.**	power = energy transferred ÷ time (P = E ÷ t)	☐
15	**State the equation linking power, work done and time.**	power = work done ÷ time (P = W ÷ t)	☐
16	**In words, state the equation for efficiency in terms of energy.**	efficiency = useful output energy transfer ÷ total input energy transfer	☐
17	**In words, state the equation for efficiency in terms of power.**	efficiency = useful power output ÷ total power input	☐
18 ★	**State what is meant by dissipated energy.**	Energy transferred to the surroundings and stored in less useful ways.	☐
19	**State what eventually happens to wasted energy.**	It is transferred to the surroundings, which warm up slightly.	☐
20 ★	**Two motors lift the same load through the same height. Motor A is quicker. Which is more powerful, and why?**	Motor A. It transfers the same energy in less time.	☐
21 ★	**Explain why efficiency can never be greater than 1 (100%).**	Energy cannot be created, so useful output cannot exceed total input.	☐
22	**A motor is 30% efficient. What does this mean?**	30% of the input energy is transferred usefully; 70% is wasted.	☐

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